Nouf Alharbi

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SUMMARY

Computer Science graduate with experience in data analysis, skilled in Python, Pandas, SQL, and Excel for cleaning, analyzing, and visualizing data into reports and dashboards. Extends this into predictive modeling and data-driven applications, applying machine learning to demand forecasting and engagement prediction, with analytical thinking and a keen interest in growing as a Data Analyst.

EDUCATION

Bachelor's Degree in Computer Science, Taibah University | 08/2020 – 05/2025
GPA: 4.64 of 5 | Second Honor

- Core foundation in programming, data structures, algorithms, and databases, applied to data analysis.

EXPERIENCE

Data Science Bootcamp, WeCloudData & Saudi Digital Academy | 08/2026 – 09/2026

- Completed intensive, project-based training in data analysis, statistics, machine learning, and deep learning using Python.
- Applied data preparation, feature engineering, predictive modeling, and evaluation to real-world datasets (MORROW, TikTok).

Software Development Bootcamp (Python), WeCloudData & Saudi Digital Academy | 05/2026 – 07/2026

- Completed intensive training in Python application development: backend APIs, web interfaces, and relational databases.
- Built data-driven applications (FLOWIN) while collaborating in Agile teams using Git/GitHub and AI-assisted development tools.

PROJECTS

MORROW - Data Science Bootcamp, Saudi Digital Academy & WeCloudData | 08/2026 – 09/2026

- Built a retail demand forecasting framework across 1,728 time series using Python, XGBoost, LightGBM, and DeepAR to optimize replenishment decisions.
- Engineered temporal features (lags, rolling stats, promotions) and evaluated six models, achieving the lowest RMSLE of 0.4761 with DeepAR.
- Classified demand into three patterns to select the best model per pattern, and delivered actionable recommendations through an interactive Streamlit dashboard.

TikTok Video Engagement Prediction – Data Science Bootcamp In-Class Competition

- Built a leakage-safe ML pipeline to predict Day 30 video views using metadata and first-5-day engagement data.
- Engineered temporal and creator features with as-of merging and stratified 5-fold validation, preventing data leakage.
- Developed a two-stage CatBoost model, reducing CV RMSE from ~138,300 to ~102,942 and reaching 81,298 on the Kaggle public leaderboard.

FLOWIN - Software Development Bootcamp (Python), Saudi Digital Academy & WeCloudData | 05/2026 – 07/2026

- Built an admission data pipeline transforming raw applicant Excel records into a structured, decision-ready dataset.
- Developed FastAPI backend and React interfaces with role-based access control (RBAC) for applicant management and dashboard analytics.
- Applied score-based validation rules (English, technical, cheat-score) to automate Accepted / Backup / Rejected shortlist decisions.

IoT-Based IV Monitoring System (IOT-MSIIA) - Graduation Project, Taibah University | 09/2024 – 05/2025

- Developed an IoT-based IV monitoring system using Arduino, ESP8266, and Firebase for real-time tracking and alerts.
- Built an Android app with Firebase to display live data, running a usability study with 97 participants to validate requirements.
- Achieved ~95.6% detection accuracy across 50+ trials with <5s latency, and 100% packet delivery across 20 integration tests.

CERTIFICATIONS

PCEP™ — Certified Entry-Level Python Programmer, Python Institute | 07/2026

TECHNICAL SKILLS

Data Analysis — Pandas, NumPy, Excel, OpenPyXL, Matplotlib, Streamlit Dashboards
Analytical Methods — Data Cleaning & Validation, EDA, Data Visualization & Reporting, Feature Engineering, Time-Series Forecasting
Data & Databases — SQL, SQLite, Firebase Firestore
Machine Learning & Deep Learning — Scikit-learn, XGBoost, LightGBM, CatBoost, DeepAR, PyTorch
Programming — Python, Java
Web Development — FastAPI, REST APIs, SQLAlchemy, React, HTML, CSS, JavaScript
Tools — Git, GitHub, Docker, Streamlit
Soft Skills — Analytical Thinking, Problem Solving, Attention to Detail, Teamwork, Time Management

LANGUAGES

Arabic, English